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Emittance compensation in a return arc of an energy-recovery linac

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Abstract

Dilution of the electron beam emittance through a return arc is one of the critical problems in the design of an energy-recovery linac for a high-power FEL and synchrotron light sources. This study reports that the emittance dilution can be compensated by matching a beam envelope to the CSR-induced dispersion. In linear regime, this optimized design of an achromatic cell is obtained from a simple matrix calculation.
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1. Introduction

An energy-recovery linac (ERL) is considered to be a promising device for a high-power free-electron laser (FEL) [1] and a next-generation synchrotron light source [2]. Since the generation of electron bunches with durations shorter than picoseconds is an essential requirement for such devices, emittance dilution arising from coherent synchrotron radiation (CSR) is one of the critical issues in the design of ERLs.

For an electron bunch of Gaussian temporal distribution, rms energy spread caused by CSR is

$$\Delta E_{\mathrm{rms}} = 0.2459 \frac{eQL_{\mathrm{b}}}{4\pi\varepsilon_0 \rho^{2/3} \sigma_{\mathrm{s}}^{4/3}}, \quad (1)$$

where Q is the bunch charge, L_{b} is the bending path length, ρ is the bending radius, and σ_{s} is the rms bunch length. This energy spread induced in an achromatic cell results in the growth of projection emittance at the cell exit. It is known that this emittance growth can be compensated by setting the cell-to-cell betatron phase advance at an appropriate value [3].

The present paper reports that the emittance compensation can be achieved in a single achromatic cell by matching a beam envelope to the CSR-induced dispersion.

2. Weak CSR, linear analysis

When the electron energy E_0 is much larger than the CSR-induced energy spread, $\Delta E/E_0 \ll 1$, a linearized approximation can be adopted to solve

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the electron dynamics. The CSR-induced energy spread results in the displacement of bunch slices in (x, x') phase space at the end of an achromatic cell. In the linear regime, it can be assumed that all the bunch slices align on a single line as shown in Fig. 1, which also indicates that the projection emittance depends on the orientation of the CSR kick and the phase ellipse, and has a minimum value, when the achromatic cell is designed so that the CSR kick coincides with the orientation of the phase ellipse at the cell exit. This study shows that an achromatic cell for such minimum emittance can be designed by a simple matrix calculation.

A first-order equation of electron motion in a uniform field of a dipole magnet is

$$x'' = -\frac{x}{\rho^2} + \frac{1}{\rho}(\delta_0 + \delta_{\mathrm{csr}} + \kappa[s - s_0]), \tag{2}$$

where the variable x denotes the deviation of an electron from the ideal path in the bending plane, x'' is the second-order derivative of x with respect to a coordinate along the ideal path s, ρ is the bending radius, and δ_0 is the initial momentum deviation at $s = 0$ normalized by the reference momentum. The last two terms on the right-hand side have been added to calculate the CSR effect: δ_{csr} is the normalized momentum deviation caused by CSR in the upstream path ($0 < s < s_0$), $s = s_0$ is the entrance of the bending magnet, $\kappa = W/E_0$ is the normalized CSR wake potential in the bending path determined by CSR wake potential W and the reference energy E_0.

Although the wake potential is a function of the bending radius, the bunch profile, and the position of an electron in the bunch, in the following discussion, it was assumed that each electron experienced constant CSR wake through the entire bending path. This assumption is valid under the following conditions: (1) all the dipole magnets have the same bending radius, (2) the electron bunch does not change its longitudinal profile, and (3) the transient CSR effect at the entrance and exit of the magnet is not large. The first two conditions are reasonable for a return arc of high-energy ERLs, but the last condition is delicate to apply. The effect of transient CSR will be examined later in a numerical example.

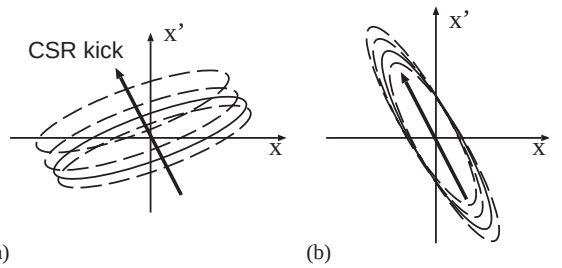


Fig. 1. Electrons in the (x, x') plane and displacement of beam slices due to the CSR kick. The emittance growth depends on the CSR kick direction and the phase ellipse orientation.

In the constant CSR wake regime, Eq. (2) can be solved analytically, and electron dynamics through an achromatic cell is calculated using the extended R-matrix [4]. A vector was defined to express electron motion: $\vec{x} = (x\ x'\ \delta_0\ \delta_{\mathrm{CSR}}\ \kappa)^{\mathrm{T}}$ and an R-matrix for a sector magnet

$$R_{\mathrm{bend}} = \begin{pmatrix} \cos\theta & \rho\sin\theta & \rho(1-\cos\theta) & \rho(1-\cos\theta) & \rho^2(\theta-\sin\theta) \\ -\frac{1}{\rho}\sin\theta & \cos\theta & \sin\theta & \sin\theta & \rho(1-\cos\theta) \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & \rho\theta \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}, \tag{3}$$

and similar R-matrices for a drift and a quadrupole, by which the electron motion through a beam transport can be tracked,

$$\vec{x}(s_1) = R_{0\to 1}\ \vec{x}(s_0). \tag{4}$$

Following the momentum dispersion function to calculate the effect of initial momentum error, (η, η'), the CSR wake dispersion function, (ζ, ζ'),

defined below was introduced:

$$(\zeta_x(s_1)\ \zeta'_x(s_1)\ 0\ L_b(s_1)\ 1)^T = R_{0\to 1}\ (\zeta_x(s_0)\ \zeta'_x(s_0)\ 0\ L_b(s_0)\ 1)^T \quad (5)$$

where $L_b(s_1)$ is the total bending path length for $0<s<s_1$.

Using the CSR wake dispersion function, it is possible to calculate the displacement of electrons in the (x, x') phase space, and the emittance growth due to the CSR effect. Unnormalized emittance at an arbitrary position along the beam transport is obtained by

$$\varepsilon^2 = (\varepsilon_0\beta_x + D^2)(\varepsilon_0\gamma_x + D'^2) - (\varepsilon_0\alpha_x - DD')^2 \quad (6)$$

where $(\alpha_x, \beta_x, \gamma_x)$ are Courant–Snyder parameters, ε_0 is the initial emittance and (D, D') is the rms spread of the CSR wake dispersion obtained by

$$(D, D') = \Delta\kappa_{rms}(\zeta_x, \zeta'_x). \quad (7)$$

$$\Delta\kappa_{rms} = \Delta E_{rms}/E_0/L_b. \quad (8)$$

Terms related to the initial momentum error have been omitted from Eq. (6), because the emittance is usually evaluated where the momentum dispersion is zero, $\eta = \eta' = 0$.

As an example of the linear analysis, the emittance growth in a triple-bend achromatic cell (TBA) was calculated: bending radius $\rho = 25$ m, bending angle $\theta = 3 + 6 + 3 = 12^\circ$, and the quadrupoles inside the cell were chosen as isochronous parameters. Fig. 2 shows the betatron envelope, the momentum dispersion and the CSR dispersion functions for the bending plane, in which the strength of the quadrupoles outside the cell has been determined so that β_x and ζ_x have same envelope after the cell. This matched envelope corresponds to the coincidence of the CSR kick with the beam ellipse in the (x, x') phase space, which results in the minimum emittance.

The emittance compensation in the optimized TBA cell has been confirmed by particle tracking code ELEGANT including CSR calculation [5]. The initial electron bunch parameters were assumed to be as follows: central energy $E_0 = 3.07$ GeV, bunch charge $Q = 770$ pC, normalized rms emittance $\varepsilon_{n,0} = 0.1$ mm-mrad, longitudinal size $\sigma_s = 30$ μm (100 fs), uncorrelated energy spread

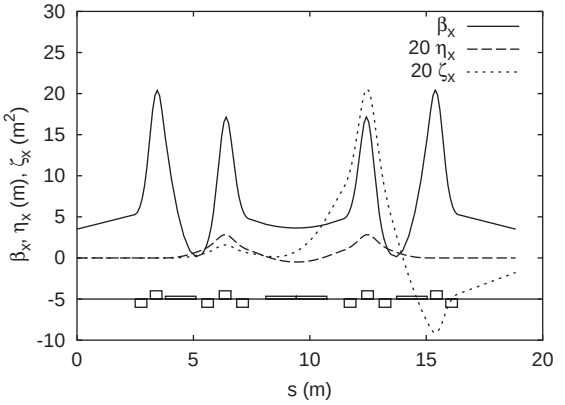


Fig. 2. Betatron function β_x, momentum dispersion η_x and CSR wake dispersion ζ_x of an optimized triple-bend achromatic cell.

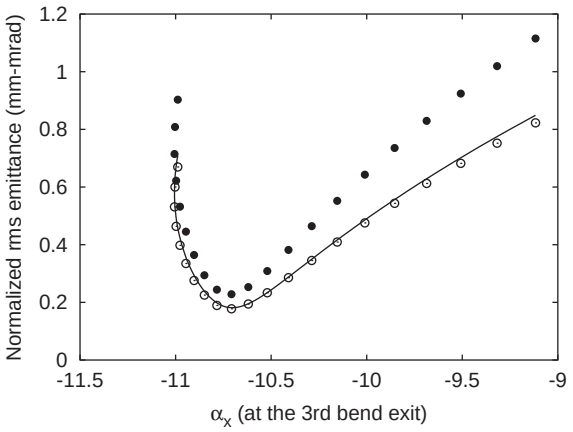


Fig. 3. Emittance growth as a function of α_x at the 3rd bend exit. First-order matrix (solid line), particle tracking without transient CSR (○), particle tracking with transient CSR (●).

$\sigma_E/E_0 = 0.02\%$, and Gaussian distribution of electrons in the 6D phase space. Projection emittance at the cell exit is calculated with scanning Courant–Snyder parameters at the cell entrance: $-5.0<\alpha_x<3.0$, $\gamma_x = 0.29$ m^{-1}.

Fig. 3 shows the calculated projection emittance as a function of α_x at the third bend exit. It has been confirmed that the projection emittance is a function of the beam envelope inside the TBA cell and can be estimated by matrix calculation. The matrix approach gives an optimized design for emittance compensation even when the transient CSR is included.

3. Strong CSR, nonlinear regime

When the CSR-induced energy spread is relatively large, higher-order terms should be taken into account. It was observed that the emittance growth due to the CSR in the nonlinear regime was related to the phase ellipse parameters at the arc exit as well as the linear analysis.

Emittance growth in the JAERI-ERL arc is calculated, where an electron bunch, $E_0 = 17$ MeV, $Q = 0.5$ nC, $\sigma_s = 660$ μm (2.2 ps), $\sigma_E/E_0 = 0.2\%$, and $\varepsilon_n = 20$ mm-mrad, travels through a triple-bend arc, $\rho = 0.23$ m, $\theta = 60 \times 3 = 180^\circ$. The CSR-induced energy spread after the arc was found to be 38 keV (0.22%) from Eq. (1).

From the simulations by ELEGANT, it can be seen that the emittance at the arc exit is a function of the beam envelope inside the arc. Figs. 4 and 5 show the optimized beam envelope for the minimum emittance growth and the emittance values for various beam envelope, respectively. In the emittance calculation, the ellipse parameters at the arc entrance were scanned: $-2 < \alpha_x < 2$, $\gamma_x = 3.6\ \mathrm{m}^{-1}$. The emittance growth by the second-order aberration for the initial rms energy spread, $\sigma_E/E_0 = 1\%$, was also plotted, where the CSR effect was absent.

Fig. 5 shows that the emittance growth due to the transient CSR in the JAERI-ERL arc cannot be estimated by linear matrix analyses. This is because the horizontal beam size is not small enough compared with the bending radius. Moreover, the CSR-induced energy spread is also relatively large, thus, the first-order approximation is not adequate in this case. However, there exists an optimum beam envelope for the minimum emittance growth. Further investigations on the correlation between the CSR-induced emittance growth and the second-order aberration shown in Fig. 5 should be conducted in future.

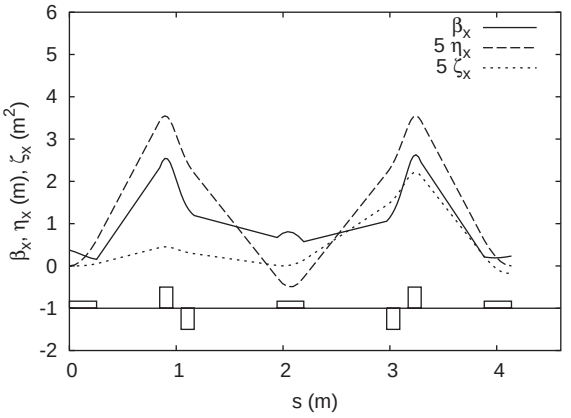


Fig. 4. Betatron function β_x, momentum dispersion η_x, and CSR wake dispersion ζ_x of the optimized JAERI-ERL arc.

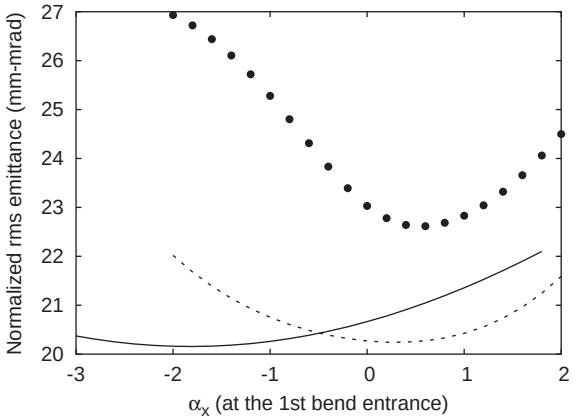


Fig. 5. Emittance growth as a function of α_x at the arc entrance. First-order matrix (solid line), particle tracking with transient CSR (•), Second-order aberration for large initial energy-spread, $\sigma_E/E_0 = 1\%$, without CSR (dashed line).

4. Conclusions

This study shows that CSR-induced emittance growth in an achromatic cell of an ERL return arc can be minimized by choosing an appropriate beam envelope inside the cell. The optimum cell design is derived from matrix calculations in the linear regime, and from particle tracking in the nonlinear regime.

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